CO 4 - Assessment Tool
Comparative Analysis

1) Heap vs Sorted Array:-

A System that repeatedly inserts elements and retrieves the highest-priority element is an priority queue.

Operation	Heap	Sorted Array
Insertion	O(log n)	O(n)
Retrieve highest	O(1)	O(1)
Deletion of highestes priority	O(log n)	O(1)
overall efficiency	high	Lower

Efficiency in Real-time System:-

• Heap:- Efficient when element are Continuous inserted and highest-priority element must be processed quickly.

Sorted Array:- Retrieval is fast, but insertion requires shift element, making it inefficient when updates are frequent.

2) Linear Probing vs separate Chaining

Factors	Linear Probing	Separate Chaining
clustering effect	High; suffers from primary clustering	very low; no primary clustering
memory usage	Lower; stores element	Higher, Dynamic
High load factor	Performance decrease	Generally better
Insertion	can become slow	efficient
Deletion	more complicated	Relatively easy
Implementation	simple	slightly more complex

a) Clustering effect:-

→ Linear Probing:- When collisions occur, element are placed in consecutive empty positions, This create clusters, increase search time.

→ Separate Chaining:- Colliding elements are stord in a separate chain at the same index, so clustering is much less promblematic.

B) Memory Usage:-

→ Linear Probing:- Uses less memory because all the elements are stored directly in the hash table.

→ Separate Chaining:- Require extra memory for linked list or other dynamic structures.

2/11/20